INDIVIDUAL PROJECT REPORT

Review 1 — Problem Identification & Proposal (24/8/2026)

AI-Driven Edge–Cloud Network for Cyclone Prediction and Resilient Ocean Telemetry

Student	Register No.	Team No.	My role
Anvay Sharma	24BCE0824	_____	Edge gateway and edge intelligence (classification, satellite image analysis, ML models)

Team: Laksh Ahuja (24BDS0212), Anvay Sharma (24BCE0824), Sreehari (24BDS0417). **Repository:** https://github.com/laksh-ahuja06/AI-DRIVEN-EDGE-CLOUD-NETWORK

Summary

Our project is a cyclone-monitoring network that must work over slow, lossy links. In this report I explain the problem from the point of view of the network edge: deciding *at the gateway* which data is urgent, and reducing heavy satellite images to the information that actually matters before they consume the backbone. I present the evidence, the literature I surveyed, our proposed solution and my plan.

1. Problem Statement

Edge devices in a cyclone-monitoring network forward everything they receive, in arrival order and at full size, so a multi-megabyte satellite image or a routine reading can delay the one packet that shows a storm is intensifying.

- **Why it matters:** the earlier a dangerous trend is recognised, the more time there is to warn and evacuate.
- **Who is affected:** fishing communities and coastal residents, and meteorological and disaster agencies that depend on near-real-time data.
- **If unsolved:** a threshold-only system gives **0 s** of warning before a threshold is crossed, and one raw 2 MB image saturates a 9.6 kbps link for about half an hour.

Objectives of My Part

Objective	Measurable success criterion
Classify every packet at the gateway as CRITICAL or NORMAL	No breach of 60 km/h / 990 hPa missed
Warn before thresholds are crossed using the trend of readings	Positive lead time in test scenarios
Cut satellite data sent over the backbone	> 95 % fewer bytes than the raw frame
Gateway that never crashes on bad input	Malformed packets counted and dropped
Fair, repeatable evaluation of every model	Held-out scenarios and a simple baseline for each

2. Evidence the Problem is Real

Table 1: Evidence that the problem is real

Source	Published fact / figure	Why it matters for my part
WMO Atlas 1970–2019 [1]	Storms caused about 577,000 deaths worldwide in 1970–2019 and were the most common cause of economic damage among weather hazards.	The cost of recognising danger late is very high, so even minutes of extra warning matter.
Chen et al., 2019 [2]	The Dvorak method is subjective and gives inconsistent estimates; their CNN needed 1,097 storms of training data and reached an 8.39 kt error.	Accurate image analysis exists but is heavy and central; the edge needs a light version.
Engineering estimate (our calculation)	A 4-band 16-bit 512 × 512 frame is 2,097,152 bytes; a record of about 200 bytes of storm features is roughly 10,000 times smaller.	Sending features instead of images is what makes satellite data usable on a slow link.

3. Literature Survey

Table 2: Literature survey (15 articles, 2018–2022)

#	Author / Year	Method used	Result	Limitation
1	Olander & Velden, 2019 [3]	Advanced Dvorak Technique update	Objective intensity from IR images	Runs centrally on full images
2	Pradhan et al., 2018 [4]	Deep CNN on IR imagery	Learned intensity estimation	Needs GPU and labelled data
3	Chen et al., 2019 [2]	CNN on IR + microwave channels	8.39 kt error on 2017 storms	Heavy model; central
4	Wimmers et al., 2019 [5]	Deep learning on passive	Intensity from microwave	Depends on polar passes

#	Author / Year	Method used	Result	Limitation
		microwave	images	
5	Rüttgers et al., 2019 [6]	GAN on satellite image sequences	6-h track forecast, ≈96 km error	Large training set; GPU
6	Denby & Lucia, 2020 [7]	Orbital edge computing	Process imagery where captured	Space segment, not ground
7	Giuffrida et al., 2022 [8]	On-board DNN cloud screening	Discards cloudy frames, saves downlink	Cloud screening only
8	Chen & Ran, 2019 [9]	Review of deep learning at the edge	Techniques to fit DL on edge devices	Mostly video, not sensors
9	Wang et al., 2020 [10]	Edge computing + DL survey	Inference and training at the edge	No disaster use case
10	Ray, 2022 [11]	Review of TinyML	ML on microcontrollers	Tools still immature
11	Banbury et al., 2021 [12]	MLPerf Tiny benchmark	Fair comparison of tiny models	Benchmark, not application
12	Van Houdt et al., 2020 [13]	Review of the LSTM model	Formulation, training, uses	Generic; framework-based
13	Lim & Zohren, 2021 [14]	Survey of deep time-series forecasting	Compares RNN, attention, hybrids	Not edge-constrained
14	Rudin, 2019 [15]	Case for interpretable models	Simple models for high-stakes use	Position paper
15	Grinsztajn et al., 2022 [16]	Benchmark: trees vs DL on tables	Simple models often win on tabular data	No time-series focus

Gap: image-based cyclone analysis (ADT, CNNs, GANs) runs centrally on full images, and on-board satellite AI so far only screens out cloudy frames. Edge-AI reviews show how to fit models on small devices but not for disaster telemetry, and interpretability work argues for simple models where lives are at stake. Nobody we found extracts storm features at the *ground station* and sends only those, or uses the *trend* of a sensor's readings at the gateway to raise priority before a threshold is crossed. These two ideas are my contribution to the project.

Foundational methods we also build on (not counted in the 15): the Dvorak cloud-pattern technique [17] and the original LSTM [18].

4. Proposed Solution Form

Form: coding and modelling. The edge gateway and the edge-AI functions are software by nature. We will first build a rule-based gateway (thresholds on wind and pressure) so the pipeline can be verified, then add lightweight models that can run on a laptop or a Raspberry Pi. A hardware-first approach was rejected because the decisions at the edge must be tested on many storm scenarios, which only simulation provides.

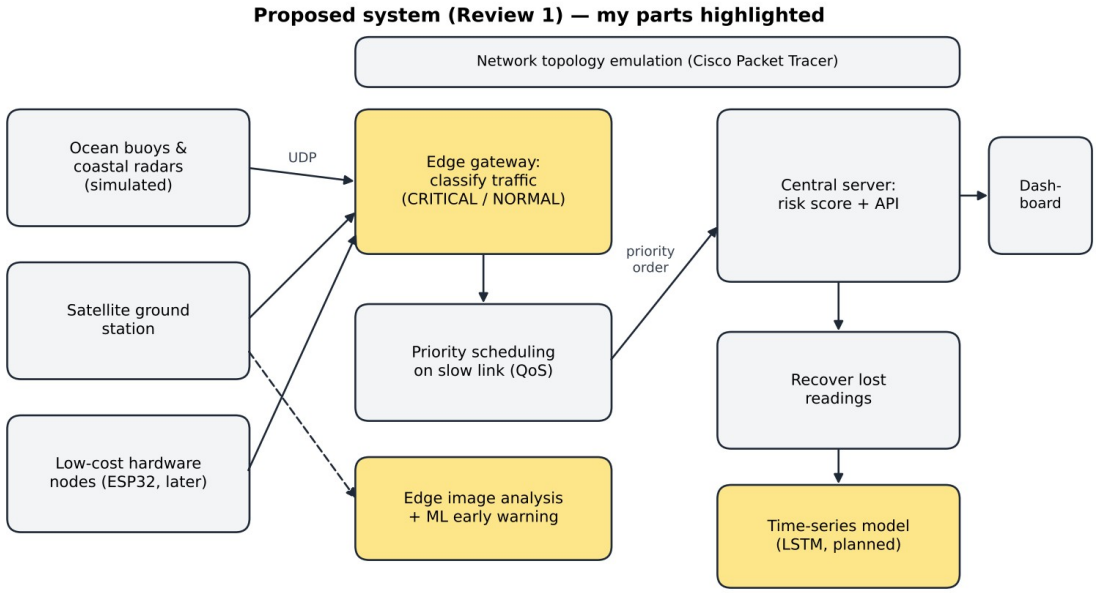


Figure 1: Proposed block diagram; my parts (gateway classification, edge image analysis, ML / LSTM) are highlighted

Planned Method for My Part

Component	Planned method
Gateway	Two threads — receiver and forwarder — joined by a priority queue; threshold rule first so the pipeline can be verified.
Early warning	Features from the last few readings (pressure falling, wind rising, distance to the thresholds) fed to a small, explainable classifier.
Satellite images	Extract the cues forecasters use — cold cloud tops, symmetry, an eye — at the ground station and send only those numbers.
Evaluation	Compare every model with a simple baseline on scenarios it has not seen.

5. Initial Technical Feasibility

Table 3: Feasibility of my part

Area	Status
Skills I have	Python, NumPy, basic machine learning (regression, classification), threading
Skills I must learn	Image processing on multi-band data, recurrent networks (LSTM) and how to train them, evaluating models fairly (held-out data, baselines)
Tools needed / access	Python, NumPy, Matplotlib (free); laptop CPU is enough for small models; real satellite imagery from public portals (MOSDAC) later
Risks	No labelled satellite data (mitigation: synthetic frames with known intensity); ML may raise false alarms (mitigation: keep the threshold rule as a safety floor); deep-learning frameworks may be too heavy for edge devices

Table 4: Risk register for my part

Risk	Likelihood	Impact	Mitigation
No labelled satellite images	High	High	Synthetic frames with known storm intensity
ML classifier raises false alarms	Medium	Medium	Keep the threshold rule as a safety floor
Deep-learning frameworks too heavy for the edge	Medium	Medium	Prefer small NumPy models
Gateway crashes on bad packets	Medium	High	Validate every packet before use

6. Team Role Split and Timeline to Review 2

Table 5: Team role split and interfaces

Member	Responsibility	Interface with the gateway (my module)
Anvay Sharma (24BCE0824) — me	Edge gateway, classification, satellite analysis, ML models	Receive, check and label every packet; decide what leaves the edge first; shrink satellite data
Sreehari (24BDS0417)	Sensor and satellite simulators, hardware nodes, gap recovery	Upstream of me: sends JSON readings to my UDP port 9001
Laksh Ahuja (24BDS0212)	Prediction server, dashboard, link QoS, testing	Downstream of me: my forwarder sends to his server on UDP port 9002

My role is the intelligence at the edge: the gateway that classifies and forwards each packet, and the models that let it recognise danger early and shrink satellite data.

Table 6: My weekly plan from Review 1 to Review 2

Week	Dates	My milestone	Deliverable / check
1	25–31 Aug	Define CRITICAL / NORMAL classes and thresholds with the team; design the gateway threads (receiver + forwarder)	Design note and agreed record format
2	1–7 Sep	Implement the gateway: UDP receive, threshold classification, priority queue	Gateway forwards packets in priority order
3	8–14 Sep	Alert mode (throttle normal traffic during an alert); integration with sensors and server; literature study for the ML classifier	Gateway shown in the Review 2 demo

R1 Video Plan

Segment	What I show
Problem and evidence (1.5 min)	Face + screen; the WMO loss figures and why late detection is costly
Literature gap (1.5 min)	Face + screen; papers on satellite intensity estimation and edge AI; the edge gap
Proposed solution and feasibility (2 min)	Face + screen; the gateway and edge-analysis blocks I will build
My role and timeline (1 min)	Face + screen; my three-week gateway plan

References

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